

Experiment 1 - Butterworth IIR Filters

1.1. Design and Analysis of a Butterworth IIR Low-Pass Filter Using MATLAB.

Program:

EXP 1.1

```
clc;  
clear;  
close all;
```

```
Fs = 10000;  
Fp = 1000;  
Fst = 1500;
```

```
[N,Wn] = buttord(Fp/(Fs/2),Fst/(Fs/2),1,60);  
[b,a] = butter(N,Wn);
```

% Frequency Response

```
figure;  
freqz(b,a);
```

% Impulse Response

```
[h,n] = impz(b,a,50);  
figure;  
stem(n,h,'filled');  
grid on;  
xlabel('Samples');  
ylabel('Amplitude');  
title('Impulse Response');
```

% Step Response

```
figure;  
stepz(b,a);
```

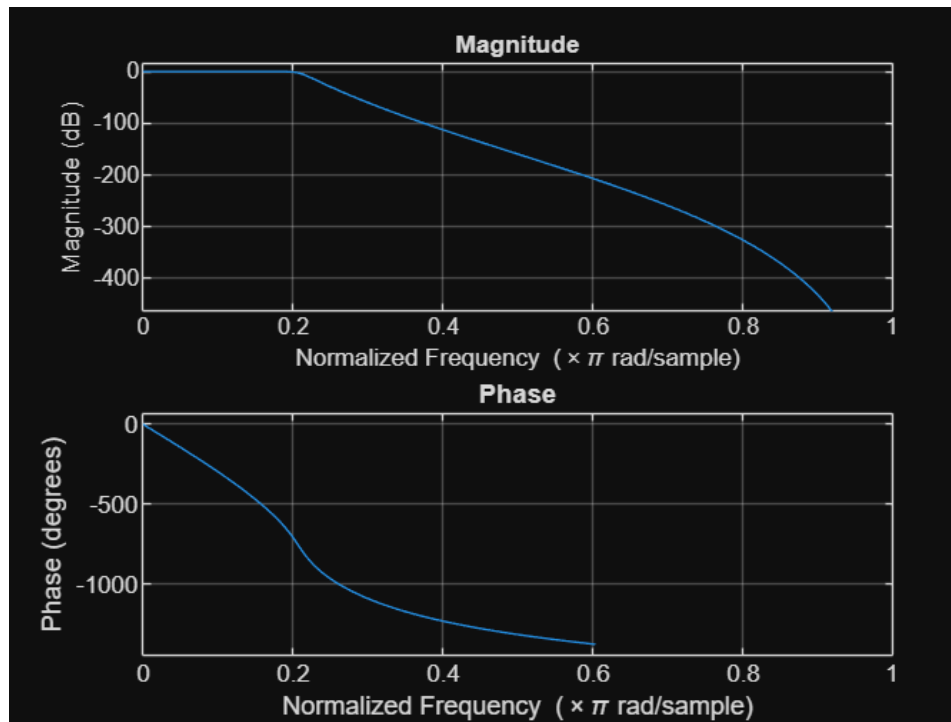
% Pole-Zero Plot

```
figure;  
zplane(b,a);
```

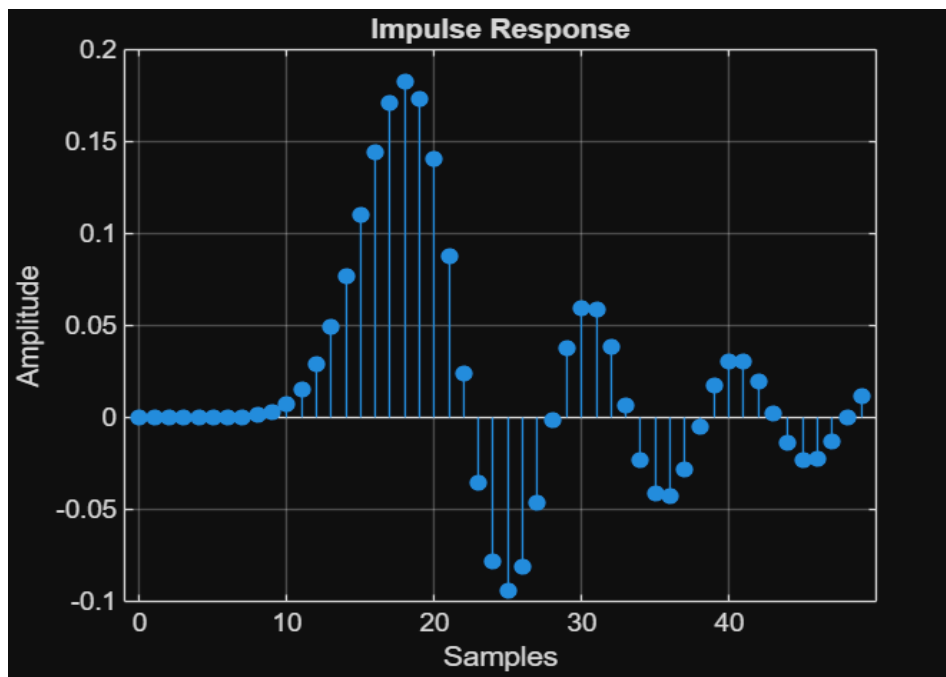
```
fprintf('Minimum Filter Order (N) = %d\n',N);  
fprintf('Cutoff Frequency (Wn) = %.4f\n',Wn);
```

Output:

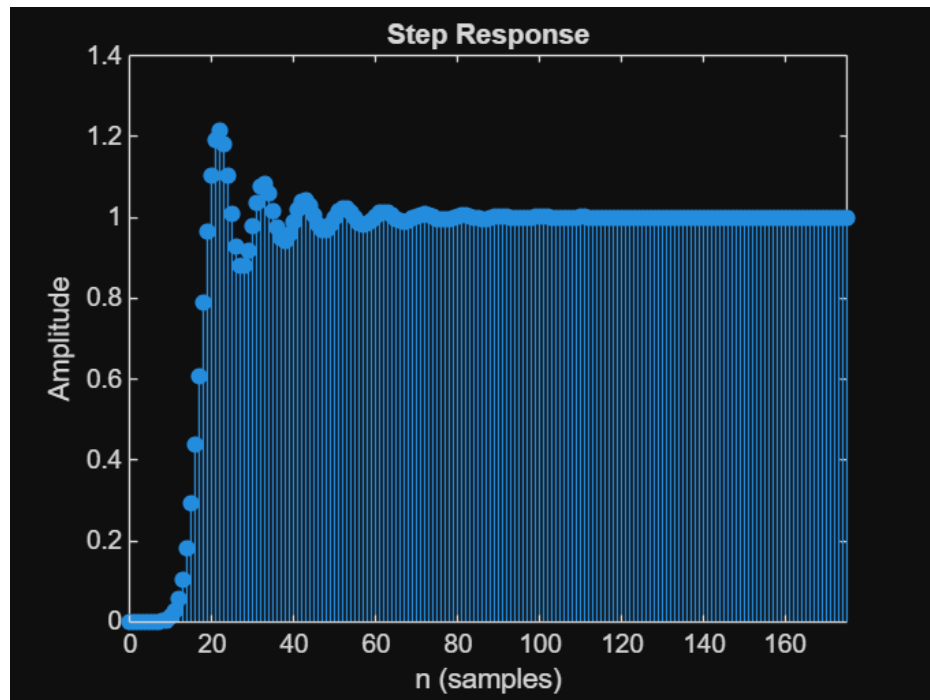
1.



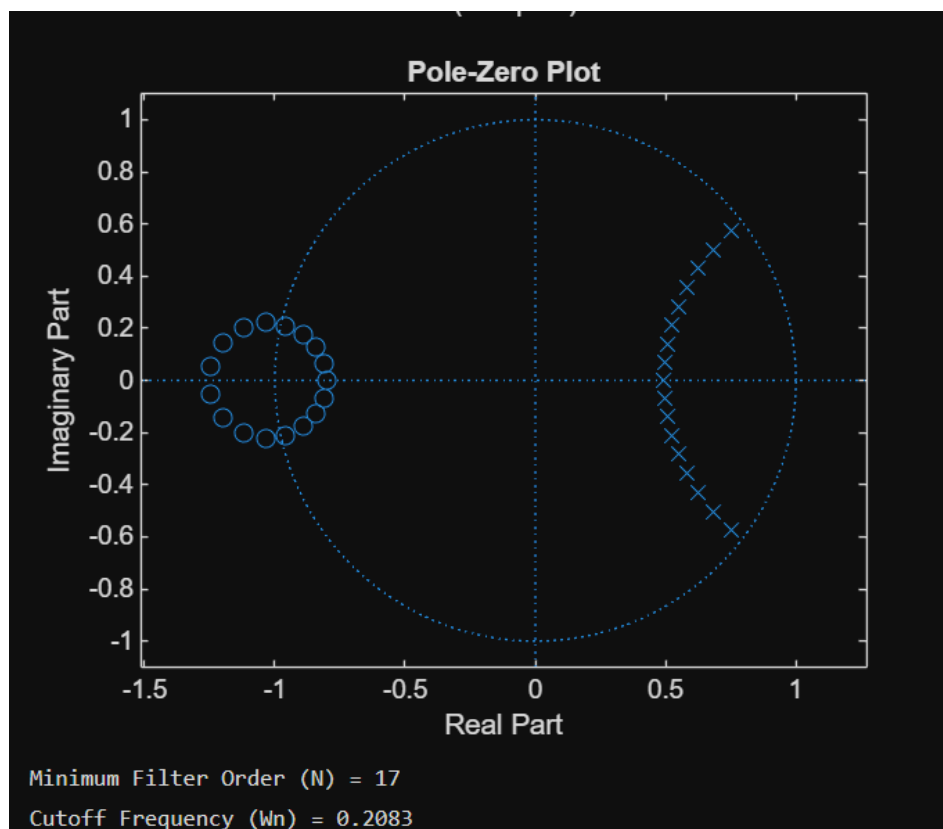
2.



3.



4.



1.2. Design and Analysis of a Butterworth IIR High-Pass Filter Using MATLAB.

Program:

```
clc;
clear;
close all;

%% EX NO: 2
% Design and Performance Evaluation of a
% Chebyshev Type-I IIR Low-Pass Filter

%% Filter Specifications
Wp = 0.40;    % Passband edge frequency (normalized)
Ws = 0.55;    % Stopband edge frequency (normalized)
Rp = 1;       % Passband ripple (dB)
As = 40;      % Stopband attenuation (dB)

%% Determine Minimum Filter Order
[N, Wn] = cheb1ord(Wp, Ws, Rp, As);

%% Design Chebyshev Type-I Low-Pass Filter
[b, a] = cheby1(N, Rp, Wn, 'low');

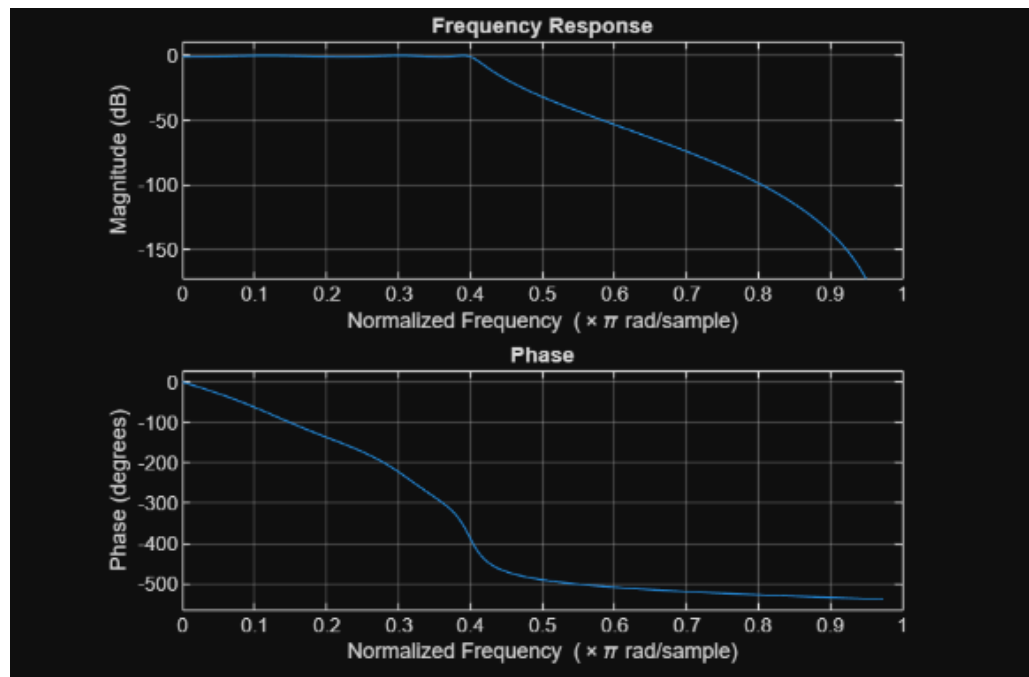
%% Display Results
fprintf('-----\n');
fprintf('Chebyshev Type-I IIR Low-Pass Filter\n');
fprintf('-----\n');
fprintf('Filter Order      : %d\n', N);
fprintf('Cutoff Frequency   : %.4f × pi rad/sample\n', Wn);

%% Stability Check
if all(abs(roots(a)) < 1)
    disp('System is STABLE');
else
    disp('System is UNSTABLE');
end

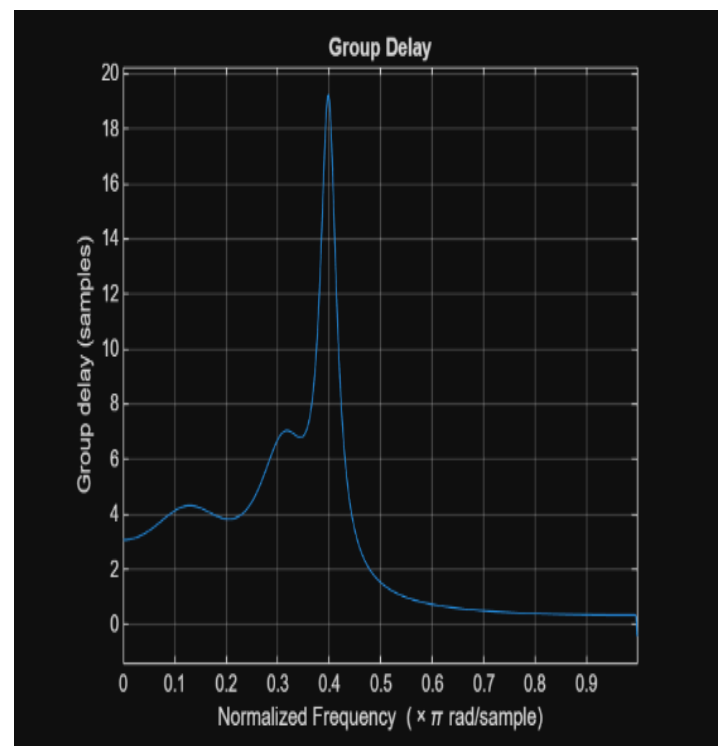
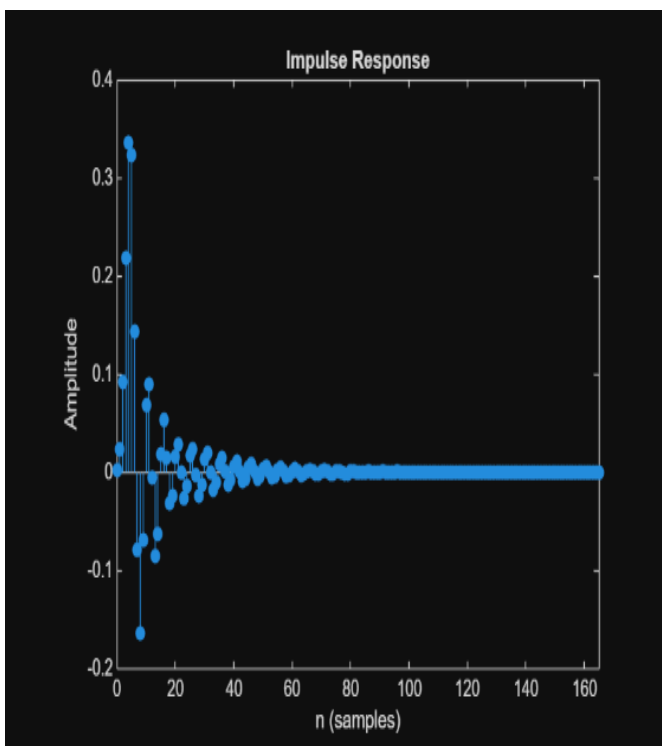
% ----- GRAPHS -----
figure; freqz(b,a); % Magnitude + Phase
figure; impz(b,a); % Impulse Response
figure; grpdelay(b,a); % Group Delay
figure; zplane(b,a); % Pole-Zero
```

Output:

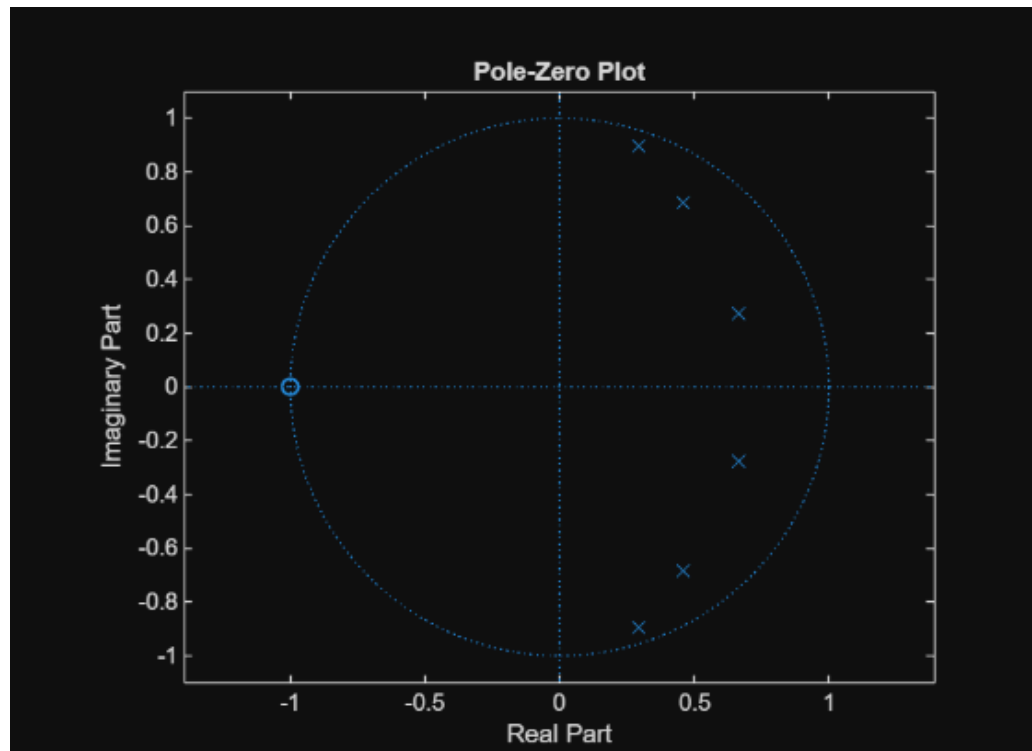
1.



2.



3.



4.

```
-----  
  
Chebyshev Type-I IIR Low-Pass Filter  
  
-----  
  
Filter Order      : 6  
  
Cutoff Frequency  : 0.4000 x pi rad/sample  
  
System is STABLE
```